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An abstract, textured background image with vibrant colors including teal, blue, orange, and yellow, resembling a close-up of a mineral or biological structure.

USING MACHINE LEARNING TO CONNECT BRANDS WITH INFLUENCERS

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Abstract

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Using Machine Learning to Connect Brands with Influencers

by Jonathan HEDLUND

With the increase of social media users, marketing through social media channels is becoming more and more important for large and small businesses around the globe. This form of marketing can be done in different ways. For example, through ads on social media platforms or by using influential opinion leaders (influencers) to reach out to a large number of customers. It can be difficult, especially for smaller brands, to navigate through the vastness of influencer marketing and find the right partner. This master thesis investigates how a process for brands to create a deal for a product/service they want to promote and how machine learning can be applied to recommend suitable influencers. Finding the right match is necessary for brands looking to take advantage of the rise of social media marketing. The thesis includes research of potential input and output for a machine learning algorithm, suggestions regarding which machine learning type and machine learning algorithm could be used, and presents a design of the deal creation process. The master thesis is conducted with the support of a startup company called Splick. Splick connects brands and influencers with an easy-to-use platform. Collaborations through Splick can be done with the help of affiliate links that track everything from clicks to sales.

Every partnership is different since it depends on the brand, the influencer, and the type of product/service the brand is trying to promote. This makes it necessary to take a wide variety of metrics and benchmarks into consideration. The thesis found that a combination of metrics from the participants of the deal and the deal itself, metrics from various social media platforms where the influencer is active, and benchmarks from Splick's affiliate link service would all be useful. The suggested output of a machine learning algorithm is common points of interest found in partnerships with similar brands and influencers that occurred in the past and turned out to be successful. Out of the machine learning types and algorithms analyzed, an unsupervised learning approach using Neural Network-based clustering seemed to be a viable way to proceed. Further research would be necessary regarding the presented machine learning approach to create a more clear plan for development and implementation.

Sammanfattning

Med ökningen av användare på sociala medier blir marknadsföring via sociala mediekanaler allt viktigare för små och stora företag runt om i världen. Denna form av marknadsföring kan utföras på olika sätt. Till exempel genom annonser på sociala medieplattformar eller genom att använda inflytelserika opinionsbildare (influencers) för att nå ut till ett stort antal kunder. Det kan vara svårt, särskilt för mindre företag, att utnyttja kraften av influencers marknadsföring och hitta rätt partner. Denna masteruppsats undersöker en process för företag att skapa en överrenskommelse för en produkt/tjänst de vill marknadsföra och hur maskininlärning kan tillämpas för att rekommendera lämpliga influencers. Att hitta rätt partner är nödvändigt för företag som vill dra fördel av marknadsföring på sociala medier. Examensarbetet inkluderar forskning om potentiell input och output för en maskininlärningsalgoritm, förslag om vilken maskininlärningstyp och maskininlärningsalgoritm som kan användas och presenterar en design av processen för att skapa överrenskommelser med influencers. Examensarbetet genomförs med stöd av ett startupföretag som heter Splick. Splick kopplar samman företag och influencers med en lättanvänd plattform. Samarbeten genom Splick kan göras med hjälp av affiliate-länkar som spårar allt från klick till försäljning.

Alla samarbeten är olika eftersom det beror på företaget, influencern och vilken typ av produkt/tjänst företaget vill marknadsföra. Detta gör det nödvändigt att ta hänsyn till en mängd olika mätvärden och riktmärken. Avhandlingen fann att en kombination av mätvärden från deltagarna och själva överrenskommelsen, mätvärden från olika sociala medieplattformar där influencern är aktiv och riktmärken från Splicks affiliate-länktjänst skulle vara användbara. Den föreslagna outputen av en maskininlärningsalgoritm är intressepunkter som hittas i samarbeten med liknande företag och influencers som inträffat tidigare och som visade sig vara lönsamma. Av de maskininlärningstyper och algoritmer som analyserats, verkade en unsupervised learning metod med hjälp av Neural Network-based clustering vara en bra metod att gå vidare med. Ytterligare forskning skulle behövas angående metoden för maskininlärning för att skapa en tydligare plan för utveckling och implementering.

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List of Abbreviations

CTR	Click Through Rate
KNN	K Nearest Neighbors
CL	Competitive Learning
SARSA	State Action Reward State Action
Lo-Fi	Low fidelity
Hi-Fi	High fidelity

Chapter 1

Introduction

1.1 Background

Social media has changed the way that brands interact with their customers. Traditional marketing such as commercials and billboards are now joined by a new giant, influencers. In the recent decade marketing through a partnership with an internet personality is becoming a necessity for a large portion of companies. Although this subject is gaining traction, and an ocean of possibilities opens up to brands all over the world, it can be difficult for brands with a small following to navigate and find the right partnership for them [21].

We are entering an era of "big data". Every time we buy a product, visit a website, browse social media, or even when we go for a walk, we are generating data [1]. We are not only generating data, we are also consuming it. Our needs are being understood and our interests are being predicted. For example how Netflix and Spotify find music that we are more keen to enjoy. With this increase of data we need automated methods for data analysis to be able to process and detect patterns in data, and then use the uncovered patterns to predict future data or other outcomes of interest. This need of automated methods of data analysis is what machine learning provides [12].

Machine learning is often divided into three different types that all have their advantages and disadvantages, the three types are supervised learning, unsupervised learning, and reinforcement learning [12]. An introduction to the different types, and some algorithms belonging to each of them will be examined.

This master thesis will research a system for matching brands with the right influencer for them, using machine learning techniques. The deal creation process should give the brand the possibility of providing all the necessary information to be able find influencers that will result in a successful partnership. The algorithm should make decisions based on the different characteristics of the brand, the deal, and the influencer possess as well as data from similar partnerships in the past.

1.2 Problem Statement

Finding the right match between brands and influencers can prove to be a difficult task in the cluttered space of influencer marketing. When the match determines the profitability of the cooperation it is important to find the right partner for your business endeavors [11].

Recommending influencers to brands for cooperation can minimize the pain of trying to find someone yourself. It would be advantageous to recommend influencers that have a higher probability to be a successful match based on data gathered from similar deals, such as the type of influencer/brand and how the partnership turned out.

1.3 Objective

The objective of this master thesis is to make suggestions related to constructing a system for recommending influencers to brands. A design will be developed for the process of first creating a deal for the product/service the brand wants to promote and how recommended influencers should be displayed. How machine learning could be used when recommending the right influencers for the brand. How the algorithm could be designed, and which machine learning algorithm could be advantageous to use and why that is the case. The choice of algorithm will be determined by evaluating and comparing different machine learning types and algorithms. The purpose of the match is to create a successful and profitable partnership. The following questions need to be answered in this paper.

- How could a design for the brands process of creating a deal for the product/service they want to promote look? How should recommended influencers be displayed?
- What benchmarks and metrics could be taken into consideration by a potential machine learning algorithm when making recommendations based on what influencers fit with which brand?
- What determines if the partnership is a successful one?
- How should the output of the machine learning algorithm be presented?
- Which machine learning type and algorithm could be suited to solve this problem and why is that the case?

1.4 Splick

This master thesis will be conducted at, and with the support of a startup company called Splick. The researcher of this study is one of the co-founders at Splick where their main responsibility is head of technology. Splick connects brands and influencers with an easy-to-use platform. These collaborations can be done with the help of affiliate links that tracks everything from

clicks to sales. An affiliate link is a link to a brand's website that the influencer can post on their Instagram, YouTube, Snapchat, or other channels. Splick's technology integrates with the brands' e-commerce platform to track sales and click data for the influencer and brand to see. When a sale occurs the affiliates are rewarded a commission paid by the brand [19].

1.5 Use case scenario

Imagine a small business looking to take its first step in the vastness of influencer marketing. Their goal is to get their products out to the right consumers. To accomplish this the business needs to have a successful partnership with the influencers they choose to advertise through. The business creates a deal through the Splick website to sell its products. In this deal, the influencer is rewarded a commission for every product sold through a link generated by Splick. From the data that Splick has about the business, such as the type of business, where the business is based, and what type of product that is being advertised, Splick should be able to recommend influencers that are deemed to be a good fit for the business. The recommendations will be based on suggestions from a machine learning algorithm.

Chapter 2

Theoretical Framework for machine learning and influencer marketing

Presented below are the necessary theory points required to make informed decisions related to the objective of this paper. The theory of this paper introduces the reader to the basics of machine learning, different types of algorithms, and examples of how they are used. In addition to the more technical machine learning theory, an introduction to influencer marketing, and affiliate marketing is presented in this chapter.

2.1 Machine learning

We are currently in an era of big data. Big data refers to data that is so large or complex that it makes it difficult or even impossible to process using the more traditional methods [3]. Instead of spending countless hours trying to decipher something out of this data ourselves, we can take help from machine learning.

Machine learning provides automated methods of data analysis. Machine learning algorithms build models based on sample data, also known as training data. With this training data, the algorithms create associations, find out relationships, discover patterns, and more [2]. Using the patterns uncovered it is possible to predict future data or perform some other kind of decision making under some degree of uncertainty [12]. The degree of uncertainty can depend on the assumptions made, the conditions imposed on the algorithm are not clear enough, or a lengthy training process that can result in a validation failure. There are different types of algorithms within machine learning that differ in their approach, output, and intent.

2.1.1 Types of learning algorithms

Machine learning is usually divided into three main types, the supervised learning approach, the unsupervised learning approach, and reinforcement learning [12]. This section will introduce the three methods and give some concrete examples of situations where they can be implemented.

2.1.2 Supervised learning

Supervised learning is a machine learning approach where an algorithm is trained on input data that has been labeled for a particular output. The algorithm is trained until it can detect an underlying pattern and relationship between the input data and output labels. The processed data is separated into different classes. The goal is for the algorithm to be able to correctly label results when presented with data that it has not seen before [15].

Supervised learning can be split into two subcategories, classification, and regression.

Classification

Training a classification algorithm is done by providing data points with a preassigned category. The goal of the algorithm is to be able to assign classes to input data that it has never seen before. An example of classification is to predict what class an image belongs to. In these types of problems, we are interested in finding and pairing the class label of an image given to the algorithm. For example, the algorithm could be tasked with figuring out if the image is of a bicycle or a car. Depending on the number of classes, the complexity and design vary, if the number of classes are equal to two it is called binary classification, if there are more than two classes we call it multi-class classification [12]. There are various algorithms available to solve classification problems. Below are some of the more common classification algorithms:

K-Nearest Neighbors K-Nearest Neighbors (KNN) is a simple example of a non-parametric classifier. The algorithm "looks at" K points in the training set that is closest to the test input. The algorithm then counts how many members of each class are in this set and returns the fraction as the estimate [12]. This means that when new data appears it can be classified into an appropriate category by using the KNN algorithm.

Decision trees A decision tree is a set of rules organized into a flowchart. The algorithm works through the flowchart step by step. Once one decision has been made the algorithm moves on to the next rule in the flowchart. A decision tree algorithm is implemented to narrow down which decision to make using the rules that are set. An example of when a decision tree approach could be implemented to choose which products should be displayed to a customer browsing an online store. Based on previous shopping behavior it could recommend products by going through a set of rules (e.g. has the customer previously purchased a toy over £100, a more expensive product can be recommended instead of a cheaper one) [7].

Regression

Regression algorithms work a bit differently from classification, in classification the output that is generated is binary, or given a class name if there are

several [1]. Regression algorithms produce a numerical relationship between the input and output data where the output is continuous. Some examples of regression models are predicting real estate prices based on zip code, estimating the temperature for a future date, and determining how much customers would be willing to pay for a specific product based on their age [15].

Linear Regression The linear regression algorithm is used to predict output that is continuous and has a constant slope. It can be implemented in situations with a continuous range, such as sales or the price of a certain product. Linear regression can be further divided into two main categories, simple regression, and multi-variable regression. Where the difference is that simple regression uses a traditional slope-intercept form and multi-variable regression uses a more complex, multi-variable linear equation [17].

Logistic Regression A logistic regression algorithm is a statistical analysis method used to predict a value based on previous observations of a data set. The algorithm can take multiple input criteria into consideration when making decisions. The model can predict a dependent data variable by analyzing the relationship between one or several existing independent variables. This type of machine learning algorithm could be implemented to try predicting whether or not a political candidate is going to win an upcoming election. A difference between using logistic regression and linear regression is that logistic regression provides values that lie between 0 and 1 while on the other hand linear regression provides all possible outputs, including negative probabilities [17].

2.1.3 Unsupervised learning approach

The goal of an unsupervised learning approach is to find "interesting patterns" in data. The provided data sets containing data points are neither classified nor labeled. Since the machine learning algorithm is not told what kinds of patterns to look for the results for this type of algorithm are much less well-defined. There is also no obvious error metric to use, unlike supervised learning where it is possible to compare a prediction of the classification of a data set to the observed value [12]. In unsupervised learning an algorithm will group unsorted information according to similarities and differences even though no categories are provided [16].

Unsupervised learning can be applied to more complex tasks than supervised learning algorithms. It could however be more unpredictable than supervised learning. An unsupervised learning AI system might for example figure out on its own how to sort bicycles from cars but it might also add unwanted and unforeseen categories that could clutter the results [16].

Clustering

Unsupervised learning is often centered around clustering. Clustering is when collecting and grouping objects or data points that are similar to each

other and dissimilar to objects in other clusters [16]. The following algorithms are commonly used for working with clustering.

Partional clustering Inside of partial clustering, the K-means algorithm is the one that is most commonly used. By using this algorithm it is possible to classify a given data set through a certain number of predetermined clusters or "k" clusters. Every cluster is assigned a center and placed with as much space between them as possible. The algorithm associates each point to the cluster center it is closest to. After that is done, the cluster centers are calculated again, and the process repeats [13].

Some examples of where this algorithm could be implemented are grouping inventory based on sales activity and identifying groups in health monitoring.

Hierarchical clustering The hierarchical clustering algorithm can be divided into two different approaches, the divisive (top-down) and the agglomerative (bottom-up) approach. The divisive approach begins as one cluster where all of the data points are collected. Afterward, it splits into multiple clusters and the data points are reassigned to the clusters that are the closest. The algorithm would ideally continue until all of the data points have their own cluster. This requires a lot of computation and is therefore not commonly used in practice [13].

The agglomerative approach starts off by considering that each data point has its own cluster. Then it gradually merges the clusters based on the distance measured between them. The process iterates and combines the clusters to eventually form larger clusters. The algorithm would ideally continue until we are left with one huge cluster containing all of the data points [13].

Neural Network-based clustering This type of clustering method is generally based on the competitive learning (CL) model which is a neural network learning process where different neurons compete in who is allowed to learn to represent the current input. Within a CL model, a stream of input patterns to a network F1 can train the adaptive weights that multiply the signals in the pathways from F1 to a coding level F2. F2 is designed as a competitive network that decides the node which should receive the largest total input. Where the winning population then triggers associative pattern learning to update the adaptive weights. A clustering implementation based on this model commonly has prototypes corresponding to the weights of neurons. CL clustering algorithms are commonly a competitive stage that precedes each learning step and decides how much a neuron should adapt its weights to a new input pattern. The goal is to minimize the distortion or quantization error in vector quantization as much as possible [13].

2.1.4 Reinforcement learning

The third machine learning type is called reinforcement learning. This is useful for learning how to act or behave when given occasional reward or

punishment signals. For example, consider how a baby learns to walk [12]. Without knowing which actions to take the algorithm must discover on its own which action yields the highest reward by evaluating the outcome after testing. In addition to the action affecting the immediate reward, the action can also affect the next situation and all subsequent rewards [18]. This tells the algorithm to seek more long-term and maximum overall reward to achieve an optimal solution. The long-term goals help with preventing the algorithm from stalling on lesser goals. After a bit of time, the algorithm will learn how to avoid the negative and seek the positive. The most common use cases are: [4]

- Gaming
- Robotics
- Resource management
- Personalized recommendations

As long as a clear reward can be applied, reinforcement learning can be used. Although it can be difficult to deploy and for it to remain within the limits of its application. Reinforcement learning is reliant on its exploration of the environment. A more complex training environment increases the demand for time and computes the resources required for the algorithm [4].

There are several different algorithms that make up the field of reinforcement learning. The algorithms differ in their approach, some important difference are due to how they go about exploring their environment, if it requires a model or is model-free, and if they are discrete or continuous.

K-armed bandit

The K-armed bandit problem is constantly faced with the choice of k different options or actions. After each choice, it has made it receives a numerical reward chosen from a stationary probability distribution that is dependant on the action that is selected by the algorithm. The objective is to maximize the expected total reward over a certain time, for example, 100 actions. The algorithm simulates a slot machine, but it has k levers instead of one [20].

By repeated action, the algorithm works towards maximizing your winnings by concentrating your actions on the best levers. Every k action has an expected or means reward given to that action. Since it is probably not know for certain what the action values are. If the action value estimates are kept, then at any time step there is at least one action with a higher estimated value than the rest. If the algorithm chooses the action with the highest estimated value it is called greedy actions and the algorithm is exploiting the current knowledge of the values of the actions. If the algorithm instead selects one of the nongreedy actions it is exploring and it enables the algorithm to improve your estimate of the nongreedy action. Exploration may be necessary to produce the maximum total reward in the long run. Since it is not possible to both exploit and to explore any single action selection a "conflict" between exploitation and exploration occurs [20].

Q-learning Q-learning learns from actions that are outside of its current policy, e.g. taking random actions, it is therefore considered to be "off-policy". The algorithms seek the best method of achieving a defined goal by seeking the maximum reward. The algorithm updates its Q-values using the Q-value of the next state and the greedy action. It estimates the total discounted future reward for state-action pairs assuming that a greedy policy was followed even though it is not following a greedy policy [20]. This type of algorithm simplifies the analysis of the algorithm and enables early proof of convergence. The policy still dictates which state-action pairs are visited and updated. Although, the only thing required for correct convergence is that all pairs continue to be updated [20].

State-action-reward-state-action (SARSA) SARSA is slightly different from Q-learning in that it learns from actions that are inside of its current policy. This means it learns according to the current action performed from the policy currently being used. SARSA updates its Q-tables values using Q-values of the next state and the current policy's action. The algorithm estimates the return for state-action pairs assuming the current policy is still being followed. SARSA learns based on the current state, current action, reward obtained, next state, and next action. Hence the naming State Action Reward State Action [20].

2.2 Overfitting

Overfitting is a common concept in the field of machine learning. Overfitting occurs when an algorithm does a poor job in generalizing unseen data. Training data is used to teach the algorithm which patterns to learn. Unseen data is data used outside of the training phase that the algorithm sees for the first time. Overfitting often occurs when a machine learning algorithm has spent too much time training with test data. This means that when overfitting occurs, the model has learned from the "noise" instead of the "signal". The signal refers to the true underlying pattern that you wish to learn the algorithm to spot. On the other hand, noise refers to the irrelevant information and randomness that occur in a dataset. Therefore you want to avoid overfitting since it will make predictions based on noise. There is also a possibility that the opposite occurs, underfitting. This means that a model is too simple, it is informed by too few features or regularized too much, which makes it inflexible in learning from the dataset. [14]

2.3 Influencer marketing

The influencer marketing strategy is a new strategy gaining more and more traction as it is adopted by more companies. Influencer marketing could be defined as using influential opinion leaders (influencers), celebrities, or non-celebrity with a large following on social media platforms. The strategy is used in order to foster a positive attitude and behavioral responses in the

influencer's followers (consumers) regarding the brand's interests by using posts shared on such platforms. This allows the influencers and followers to participate in the co-creation of the brand image on social media [11].

2.4 Affiliate marketing

There are many different kinds of co-operations between brands and influencers. For example, there are upfront deals where the influencer is paid a fixed amount upon accepting the terms of the deal. An example of an upfront deal could be where the influencer is to advertise a product for a company. Another type of cooperation is affiliate marketing. Affiliate marketing is a prominent type of performance-based marketing where brands reimburse affiliates for each customer they refer through the affiliate's marketing effort. The key objective of affiliate marketing is to sell or promote products/services through the affiliates marketing outlets to drive web traffic to advertisers and generate transactions from online users in return for a commission [5].

2.5 Benchmarks

In order for the machine learning algorithm to make educated decisions, both benchmarks from the social media platforms and benchmarks from Splick's plugin are necessary to analyze. Presented in this section are some common social media benchmarks and benchmarks currently used by Splick's affiliate marketing plugins.

2.5.1 Social media benchmarks and measurements

The benchmarks for each of social media platforms are usually determined by the benchmarks that are available for the user to view on each social media, e.g. Instagram, Snapchat, and Youtube. Listed below are some of the more common benchmarks alongside a description for each.

Followers

Influencers' usually have a large following of people they can effect with the content they produce. This metric is very straight forward and an important one. Since more followers usually mean that they have a higher chance of reaching out to more people.

Engagement Rate

The engagement rate refers to the level of interaction with followers between content created by a user and their followers. The engagement rate gives a more accurate representation of the performance of content created by a user

rather than only looking at absolute measures such as likes, shares, and comments. The engagement rate is calculated by dividing the total engagement (e.g. likes, shares, and comments) with total followers [8] .

Impressions

Impressions is a metric that is counted each time a user's ad or post is displayed to other users on that platform. It does not matter whether or not the user that is shown the content clicks on it though, only that it shows up in their timeline [22].

Reach

Reach is a metric that counts the total number of people who see your content. It differs from the impressions in that the reach is the number of unique people who see your content where one user could have multiple impressions for a single piece of content [22].

Click Through Rate (CTR)

The CTR refers to the number of clicks advertisers receive on their ads per number of impressions. The CTR is calculated by dividing the total clicks on content by total impressions. With affiliate marketing, the click-through rate could be the number of clicks on a specific link based on the total amount of impressions.

Demographic of followers

Location, age, and gender of users' followers are all-important demographic aspects to take into consideration. Knowing the demographic of your target group greatly increases your chance of being able to choose the correct channel to advertise through.

Activity timetable

The activity timetable displays on which days and which hours a user's followers are active. This can be used to predict when to post content in order to reach as many followers as possible.

Growth

This metric displays the growth of a user's follower base. The metric can be viewed over time to see how well the user is doing in increasing their total amount of followers.

2.5.2 Splick benchmarks

Listed below are the benchmarks that Splick can evaluate using the plugin developed and connected to the affiliate link service. The benchmarks can be used on their own and together with the social media benchmarks to provide more value for the influencer and for the brand. With these benchmarks, brands can measure successful cooperation in different ways, some of the brands might only take interest in the total amount of sales made and some might be more interested in the exposure they gain from appearing on the influencer's social media platform.

Clicks

Clicks are metric which refers to the number of clicks made by users on an affiliate link provided by an influencer. The amount of clicks gives a good representation of how well the ad has interacted. This metric can be used together with some social media benchmarks to analyze how well the partnership went. For example, reach and impression can be used with clicks to calculate the click-through rate of an ad.

Add to carts

Amount of products added to the online shopping cart at an e-commerce website by users referred to the website through an affiliated link. This metric counts the total amount of products added to the shopping cart and does not include how many of these add-to-carts resulted in a purchase.

Sales

The total amount of money generated by a partnership between a brand and an influencer. This metric is important in helping to grasp how profitable the cooperation turned out to be. It can also be used to calculate the influencer's total commission earned for advertising the product on their social media platforms.

2.5.3 User measurements

An important aspect of every partnership is the type of brand/influencer and what type of product/service the deal is promoting. This section presents some aspects that could be interesting to take into consideration when making decisions based on recommending a suitable candidate.

Brand information

Information about the brand is important to gather in order to gain more knowledge when recommending a suitable influencer. Some examples of factors that could be interesting to take into consideration are brand category (e.g. fashion, entertainment, and electronics), location, size, and target audience.

Influencer information

Information about the influencers is also necessary in order to better separate and divide the available influencers into different classes. Valuable information such as where the influencer is based, gender, age, and interests could all be used as input in the machine learning algorithm.

Deal information

When a brand creates a deal on the Splick website it is prompted to enter some information about the product or service that will be promoted. This information can be used to find out which influencer could be a suitable match. Information such as what type of products/service it is, the price, and the amount of commission rewarded to the influencer could all be used.

Chapter 3

Theoretical Framework for development of the user interface

Presented below are the theory points to gain insight into the design process. Theory about the iterative process and design evaluation is presented in this chapter.

3.1 Iterative User Interface Design

Using an iterative process when designing a user interface can substantially improve usability. According to Nielsen (1993), it is recognized that it is virtually impossible to design a user interface that has no usability problems from the start. Therefore a usability engineering lifecycle should be built around the concept of iteration [9].

The iterative process involves refining a design through multiple cycles based on results from user testing and other methods of evaluation. Using the results based on the evaluation can successively remove the problems with the design, the new iteration can then go through another evaluation to confirm that the problems have been fixed and no new problems have surfaced. Conducting user tests on a small number of test subjects can result in sufficient insight into the usability problems in a given design iteration [9].

The usability of a design usually goes up for each additional iteration, until the usability potentially plateaus. Figure 3.1 shows the relationship between the number of iterations and quality of the interface [9].

3.2 Heuristic evaluation

Heuristic evaluation is the process of using guidelines to evaluate the usability of a user interface in independent walkthroughs and report issues. Evaluators can use heuristic guidelines established by experts and reveal insights about their products. Nielsen and Molich, two pioneers in web usability, created a list of ten user interface design guidelines in the 1990s. These guidelines have been adopted by some of the most successful companies, such as Apple, Google, and Adobe. Nielsen and Molich suggest using the following guidelines: [6]

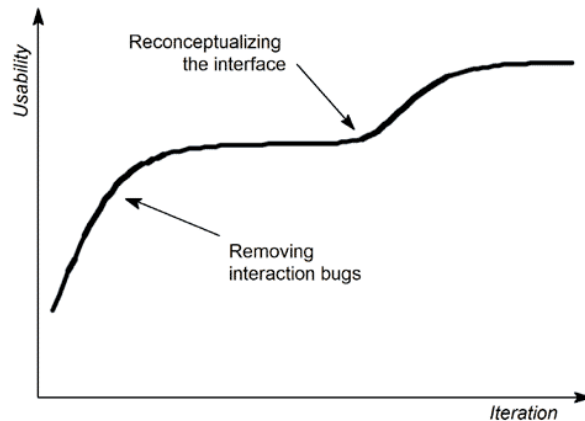


FIGURE 3.1: Interface quality as a function of the number of design iterations.

1. **Visibility of system status.** The user should always be informed of system operations with an easy to understand and visible status displayed on the screen.
2. **Match between system and the real world.** Designers should strive to mirror the language and concepts users find in the real world.
3. **User control and freedom.** The users should have the possibility to move freely in a digital space where backward steps and undoing/redoin actions are possible.
4. **Consistency and standards.** The graphic elements and terminology are consistent across the platform. For example, an icon should not represent different concepts when used on different screens.
5. **Error prevention.** Keeping the amount of errors a user can make to a minimum. The designer should eliminate or flag actions that the user can take which would result in an error.
6. **Recognition rather than recall.** Relevant instructions on how to use the system should be easily accessible for the user. The system should minimize the cognitive load of the user.
7. **Flexibility and efficiency of use.** Users should be able to customize or tailor the interface to suit their needs. Resulting in more frequent actions being achieved through convenient methods.
8. **Aesthetic and minimalist design.** The system should not include unnecessary information which compete with the user's attentions. Keep clutter to a minimum.
9. **Help users recognize, diagnose and recover from errors.** Error messages should be expressed in plain language that the user easily can understand to ensure the user knows exactly what has gone wrong.

10. **Help and documentation.** The system should provide the user with documentation that is easily accessible. Ideally the user should be able to use the system without additional documentation, but in more complex systems documentation might be necessary.

The 10 rules of thumb developed by Nielsen and Molich were created in 1990. Technology has advanced greatly since then which has made some of the guidelines from Nielsen and Molich a bit too general to evaluate the usability of designs for many of the products on the market today. Designers today should therefore be encouraged to create their own specific guidelines for their product [6].

Choosing which heuristic guidelines to use can be a difficult process. There are no fixed recommendations since each design differs. Since you have the ability to create your own guidelines the options increase. Most heuristic evaluations should involve between five and ten guidelines chosen based on the design that is being developed. Using less than five heuristics can lead to the evaluation not being able to identify potential problems. Using more than ten heuristics may overburden the evaluator [6].

3.3 Usability testing

Usability testing is one of the more popular observational methodologies to uncover problems and find solutions for user interfaces. In a usability test, a facilitator asks a participant to perform certain tasks specific to the user interface. While the tasks are being completed, the facilitator observes the participant's behavior and listens to the feedback being given [10]. There are three main parts to usability testing, the facilitator, the tasks, and the participant.

The **facilitator** guides the participants through the test process by giving instructions, answering questions from the participant, and asking follow-up questions. The main responsibility of the facilitator is to ensure that the test results in high quality, valid data, without influencing the participant's behavior [10].

The **tasks** are realistic activities that users of the interface might find themselves performing in real life. They can vary from being specific to open-ended depending on the purpose of the usability tests. How the facilitator formulates the tasks for the participant is very important. Should errors in the phrasing occur it can cause the participant to misunderstand what they're asked, it can also influence how the participants perform the task [10].

When choosing the **participants** for the usability tests, it is suggested to choose participants that are realistic users of the product/service being evaluated. However, in some instances, the participants might only share a similar background to the realistic users of the product/service. It is common for the facilitator to tell the participant to "think aloud" while completing the given tasks, the method basically means that the participant should narrate their actions and their thoughts throughout the test. This method helps the

facilitator to understand and document the participants' behaviors, goals, thoughts, and motivations [10].

It is recommended to use five participants when conducting a usability test that aims to gather qualitative data. It is proven that five participants can uncover the majority of the most common problems with the product/service [10].

Chapter 4

Methodology

4.1 Method

As mentioned in Section 1.3, the aim of this master thesis is to make suggestions related to constructing a system for recommending influencers to brands. A design will be developed for the process of first creating a deal for the product/service you want to market and how recommended influencers should be displayed. In addition to the design, which factors need to be taken into consideration when making decisions about recommending influencers to brands. What could be a suitable output for a machine learning algorithm, and which type of machine learning type and algorithm could be used for connecting brands with the right influencer for them. The method proposed in this chapter will also aim to answer the questions in Section 1.3.

4.2 Literature study

An extensive literature study to better understand the field of machine learning and influencer marketing was conducted in order to make decisions and suggestions related to the objective of the master thesis. The results of this research are provided under Chapter 2. The vast majority of the sources used in this master thesis were found using reliable sources such as Umeå University's online library and Google Scholar. In some instances where peer-reviewed sources related to a topic could not be found, websites or blog posts that seemed reliable were used.

The literature study consists of three main parts, research regarding possible inputs for the machine learning algorithm, research regarding possible output for the algorithm, and the machine learning aspect, research about which type of machine learning and which type of algorithm could be suitable.

4.2.1 Evaluating the inputs

The purpose of evaluating the benchmarks provided in Section 2.5 is to better know which factors are useful and which could be of a higher priority. What type of input the machine learning algorithm is provided is crucial to selecting a machine learning algorithm that solves the problem. With the given problem and the number of parameters available it is important to evaluate

the value they could bring towards finding a suitable match between a brand and an influencer. The following categories of inputs have been evaluated in this literature study:

- Social media benchmarks and measurements (e.g. Engagement rate, impressions, and growth)
- Benchmarks from Splick's plugin (e.g. Clicks, add to carts, and sales)
- User measurements (e.g. Brand information, influencer information, and deal information)

4.2.2 Evaluating the output

Evaluating and deciding how the output should be presented is important both for the brands that are presented with a recommendation but also for the effectiveness of the machine learning algorithm. The output of a machine learning algorithm is whatever you specify it to be. To create an effective machine learning algorithm for this scenario the information provided in this master thesis is necessary to tailor an output template.

4.2.3 Evaluating machine learning types and algorithms

Section 2.1 - 2.2 includes an extensive introduction to the field of machine learning. Three different types of machine learning, supervised learning, unsupervised learning, and reinforcement learning was introduced to gather valuable information about what their strengths are and what differentiates them from each other. In addition to the different machine learning types, a few machine learning algorithms from each machine learning type was introduced.

This was done in order to gain more knowledge about which possibilities there are for developing and implementing a solution into Splick's website. Both the different machine learning types and algorithms need to be evaluated in order to make suggestions based on where a solid starting point for developing a system for recommending influencers could be.

4.3 The design process

The design process used in this master thesis will be based on the design theory from Chapter 3. More specifically the theory about the iterative process, heuristic evaluation, and usability testing. An iterative process was used in developing prototypes for the design and the different types of evaluation methods were used to analyze the prototypes.

The design is dependent on the results regarding the benchmarks and metrics a machine learning algorithm should take into consideration when recommending influencers. Because we first need to find out which information is important in order to know how the process should be constructed and which inputs the user should be able to fill out.

4.3.1 Prototype development method

Developing a design for the recommendation process will be separated into different steps using different techniques. The approach in this thesis used an iterative process described in Section 3.1 and to create a Lo-Fi and a Hi-Fi prototype. The Lo-Fi prototype was created using pen and paper to easily sketch out ideas in a fast and efficient way. The Hi-Fi prototype was created using the user experience design tool Adobe XD. Using Adobe XD is a more time-consuming process than sketching by hand, but it results in a more realistic version that you're able to make interactive which results in the possibility of usability tests that work similarly to how a finished product would act.

4.3.2 Lo-Fi prototype

Using the results from the literature study of this master thesis as guidelines, a Lo-Fi prototype was created using pen and paper. The prototype needs to explore how the recommendation process should be constructed to make it easy for the user to navigate through it. Evaluating the Lo-Fi prototype will focus on overviews of the main components and structure of the interface, not the graphic design. Text and phrasing will not be analyzed during the Lo-Fi evaluation. Since Lo-Fi prototypes are not that interactive, a heuristic evaluation based on the theory presented in Section 3.2 was conducted. The heuristic guidelines that were chosen for evaluating the Lo-Fi prototype were:

1. **Visibility of system status.** The user should always be informed of system operations with an easy to understand and visible status displayed on the screen.
2. **User control and freedom.** The users should have the possibility to move freely in a digital space where backward steps and undoing/redone actions are possible.
3. **Recognition rather than recall.** Relevant instructions on how to use the system should be easily accessible for the user. The system should minimize the cognitive load of the user.
4. **Aesthetic and minimalist design.** The system should not include unnecessary information which competes with the user's attention. Keep clutter to a minimum.
5. **Help users recognize, diagnose and recover from errors.** Error messages should be expressed in plain language that the user easily can understand to ensure the user knows exactly what has gone wrong.

4.3.3 Hi-Fi prototype

The Hi-Fi prototype is based on the design of the Lo-Fi prototype and the heuristic evaluation. This iteration moved away from using pen and paper

and instead utilized the prototype developing tool Adobe XD. The Hi-Fi prototype simulates a more realistic design closer to a final product that users can interact with.

Evaluating the Hi-Fi prototype was done through usability tests introduced in Section 3.3. Five people participated in the usability tests. The participants were four males and one female between the age of 23 - 27. All of the participants had a lot of experience using computers but were new to the area of influencer marketing. The test gave the users a chance to explore an in-depth version with real-time interaction and feedback. The users performed a series of tasks chosen to make them interact with the whole interface. An introduction to influencer marketing and affiliate deals were given at the start of each usability test due to the difficulty of finding participants who were familiar with marketing through influencers. The tests were performed remotely via Zoom where the participants would share their screen while interacting with the prototype. The participants were asked to "think aloud" and narrate their actions and thoughts throughout the test.

The test protocol was:

1. Give the participant an introduction to influencer marketing and affiliate deals.
2. Give the user a scenario for the test. The scenario given was that the user owns an online store selling luxury watches that they want to promote using influencer marketing.
3. Instruct the participant to think out loud while performing their tasks.
4. Read each task and wait for the participant to complete it and answer potential questions that they have at the end of each task before moving on to the next one.
5. Give the participants a chance to share any final thoughts about the design.

The tasks used in the usability tests were:

1. Define the purpose of the deal by giving it a name and choosing a deal with the goal of generating sales.
2. You want to market your products in Sweden using Instagram posts and the influencer should be rewarded with a percentage rate of customers spending.
3. Provide all the information about the content of the deal before going to the next step.
4. Add Norway to the list of countries that you want to market your watches in.
5. Search new influencers and pick one of the recommended influencers you want to cooperate with.

Chapter 5

Results

In this chapter, decisions, and suggestions related to the objective of the master thesis is presented. An extensive literature study was conducted and the conclusions were based on the theory provided in Chapter 2. In addition to the literature study, user interface prototypes were developed and evaluated with the help of the design theory provided in Chapter 3.

5.1 Key aspects regarding benchmarks (Input)

There are many possible inputs for developing a machine learning algorithm suited for this scenario. In Section 2.5, the following factors were presented.

- Splick benchmarks
 - Clicks
 - Add to carts
 - Sales
- User measurements
 - Brand information (Brand cateogry, location, size, and target audience)
 - Influencer information (location, gender, age, and interests)
 - Deal information (type of products/service, price, and commission rewarded to the influencer)
- Social media benchmarks
 - Followers
 - Engagement rate
 - Impressions
 - Reach
 - Click Through Rate (CTR)
 - Demographic of followers
 - Activity timetable
 - Growth

The benchmarks that Splick can measure using the plugin connected to the affiliate link service are valuable for measuring how the partnership between a brand and an influencer actually went. Clicks, add-to carts, and sales can help with training the machine learning algorithm to be able to recommend a profitable candidate to the brand. Which benchmarks are the most important is difficult to say since different brands value different metrics and could have other motives than only maximizing sales. For example, some brands could be more interested in gaining publicity by appearing on the influencer's social media platforms.

Since this is the only data giving information about how the partnership turned out, all of these benchmarks would be necessary for the algorithm to gather as input. The theory mentions being able to combine Splick's benchmarks with some social media benchmarks to create new ways of measuring the outcome of cooperation. More research would be needed in this area to better assess possible benchmarks to incorporate.

User measurements include a lot of basic information about the participants and the deal. This information builds the foundation of the recommendation process and is therefore required to make important decisions in the initial filtering process. This information includes a wide variety of data. Partly details on the brand, such as brand category, location, and size. On the influencer side, it could include gender, age, and interests. Finally deal information such as details about the product/service, the price, geographic dependencies, and amount of commission rewarded to the influencer could all be used. This information can be used to make basic filtering. For example, brands can specify where in the world they want their products to be marketed and then influencers from that area are highly prioritized.

There are a lot of social media benchmarks available that are important to take into consideration when constructing a machine learning algorithm. The various benchmarks have their own strengths and ideally, the machine learning algorithm would take all benchmarks mentioned under Section 2.5.1 into consideration. These metrics are gathered from each social media. Which metrics are available depends from platform to platform, for example, the metrics from Instagram differ from the metrics available from TikTok. Therefore it is important to see which platforms an influencer is active on to be able to determine which benchmarks you can measure and analyze. The benchmarks previously mentioned help gauge the power to influence the influencer to possess and gives information about the influencer's followers.

Brands can measure successful cooperation in different ways, some brands might only be interesting in the total turnover of a partnership, some might be more interested in exposure and value a higher click-through rate above all. Since there are different benchmarks that could be favored above others it would be best if brands could specify which ones they value the most and the machine learning algorithm would take that into consideration when recommending an influencer. This is possible but it requires more test data and training.

5.2 Key aspects regarding what the recommendation could look like (output)

The type of output differs depending on which machine learning type and algorithm that is being used. So modelling an output before the choice has been made is not optimal. This section of the results will describe more in general what kind of output could be preferred.

The recommendations at Splick's website should give the brand a couple of influencers to choose from that the algorithm thinks could be a possible fit, based on information about the brand and what type of products/service they are promoting. The brand is then able to browse the profiles of each recommended influencer and make the final decision themselves. If the brand would prefer to browse through Splick's database of influencers and choose one without the recommendation service it should be able to do so.

The influencers that are recommended need to have an account at Splick's website. Users that were once active can stop using the site at any point. That means that it would not be beneficial to train the algorithm to only output names of potential candidates. The algorithm should instead provide suggestions on a specific category of influencers. The suggestion could be that for a specific brand an influencer with specific demographic aspects, specific interests, and a specific type of follower base could be a good match. With this type of output Splick is able to filter their influencer catalog and display influencers that satisfy the demands. The filtering would also be able to only display influencers that are active on the site to increase the possibility of a partnership being initiated.

5.3 Key aspects regarding the machine learning type and algorithms

This section of the results will go through the machine learning types and algorithms to analyze how well they could fit with the problem of recommending suitable influencers to brands. The goal of this section is to provide valuable information about where to conduct further research when starting out the development of a machine learning solution for Splick.

5.3.1 Supervised learning

Supervised learning is the first machine learning type brought up in this master thesis. In supervised learning, the algorithm is trained on input data that has been labeled for a particular output where the goal is to be able to correctly label results when presented with data that it has not yet seen before. Therefore is required to know how to classify the training data that is to be provided. Supervised learning can be split into two subcategories, regression, and classification. Regression produces a numerical relationship

between the input and output data where the output is continuous. In classification, the goal is to be able to assign classes to the input data, where the output is either a yes or no answer or given a class name if there are several.

It is difficult to label whether or not cooperation between a brand and an influencer has been successful. Especially if you only look at the benchmarks generated by Splick, since a large number of variables are in play. For example, take two partnerships with two different influencers that generated exactly the same amount of sales, is the partnership equally successful? Not really, the size of the influencer's audience and a multitude of factors are important to take into consideration to determine how well the partnership went relative to the influencer. Unless assumptions are made to manually label if cooperation turned out to be "successful" or not it will be difficult to use supervised learning in Splick's scenario, regardless of using regression or classification and their different algorithms.

5.3.2 Unsupervised learning

With unsupervised learning, the goal is to find "interesting patterns" in data. Here the provided data sets with data points are neither classified nor labeled. Because the machine learning type is not told what kinds of patterns to look for it is able to find connections that it deems to be interesting on its own. The unsupervised learning algorithms group unsorted information according to similarities and differences even though no categories are provided. Unsupervised learning is often centered around clustering data points that are similar to each other and dissimilar to objects in other clusters.

As mentioned in Section 5.3.1 it is difficult to determine if cooperation between a brand and an influencer is successful since there are a multitude of variables that needs to be taken into consideration, such as the size of the influencer, information about its followers, and information about the influencer. A machine learning type where you do not have to manually label whether or not a partnership is successful is preferable. It is also able to be applied to more complex tasks than supervised learning algorithms, which Splick's recommendation system would be.

Out of the machine learning algorithms that are mentioned under this section, Neural Network-based clustering seems to be the one that would fit best with Splick's recommendation system. The reasoning behind this is that Neural Network-based clustering can train a set of adaptive weights to update the connections between the nodes depending on which node is granting the highest output based on the input. As the algorithm receives more data it is able to give more accurate predictions on how the input should be clustered.

5.3.3 Reinforcement learning with K-armed bandit

A reinforcement learning algorithm is taught how to act or behave when given occasional reward och punishment signals. The algorithm must discover by itself which action yields the highest reward by evaluating the outcome after testing it. The action affecting the immediate reward can also affect all subsequent rewards. This tells the algorithm to seek more long-term and maximize the overall reward. After some time the algorithm is able to avoid the negative and seek the positive. Reinforcement learning can be applied as long as a clear reward can be given to the algorithm.

Reinforcement learning possesses a variety of aspects that are desired when looking at it from the view of Splick's recommendation system. Since it is hard to determine how successful a cooperation actually was it would be ideal to use an algorithm that discovers by itself which type of influencers could be a possible fit based on the brand and deal. It is possible to give occasional reward or punishment signals to the algorithm both from Splick's benchmarks but also from the participants as well. The brand or influencer could after the deal be prompted to answer how they experienced it and that could help improve the long-term effectiveness of the algorithm. Reinforcement learning would however take the most amount of data compared to the other two machine learning types to be able to start making recommendations that have some thought behind it. This means that even after implementing a reinforcement learning-based machine learning algorithm it could take months until it is able to match brands with influencers.

When developing an algorithm to maximize the expected total reward long term The K-armed bandit problem mentioned under 2.2.3 could be an interesting field to research further.

5.4 Lo-Fi prototype

This section of the results describes each part of the Lo-Fi prototype and the results from the heuristic evaluation conducted to find problems and solutions before starting the next iteration.

5.4.1 The design

As previously described, the Lo-Fi prototype was created by sketching on paper. But prior to the sketching process, the core steps of the deal creation and recommendation process in addition to what each step should consist of was listed. The reasoning behind the choice of steps and what they consist of is based on the results regarding benchmarks presented in Section 5.1. The core steps for the user to go through before they are matched with an influencer suited for their needs are presented in the list below.

1. Defining the purpose of the deal.
2. Providing information about the deal.

3. Choosing an influencer to promote the deal.

To gather all the necessary information before being able to recommend an influencer the user needs to fill out a few inputs at each step of the process. For each of the core steps the user should provide the following:

1. Defining the purpose of the deal.

- Naming the deal.
- Goal with the deal (generating sales, publicity).
- Choose where in the world you want to promote the deal (global or specific countries).
- Channels to promote the deal (Instagram, Youtube, etc.).

2. Providing information about the deal.

- Product/service.
- Type of product/service.
- Price of the product/service.
- Type of commission for the influencer.
- Amount of commission for the influencer.
- Description of the deal for the influencer.
- Media for the influencer to post.

3. Choosing an influencer to promote the deal.

- Providing recommendations for influencers through machine learning based on the input from the previous steps, information about the brand, and similar partnerships conducted on Splick's platform in the past.
- Give the opportunity for brands to search and filter all the influencers who has an active account at Splick.

Figure 5.2 below shows a close up on the "Purpose of deal" block of the deal creation process. The block contains all of the inputs listed above for this section.

A hand-drawn sketch of a form titled "PURPOSE OF THE DEAL". The form is enclosed in a rectangular border. At the top, the title "PURPOSE OF THE DEAL" is written in a large, bold, sans-serif font, with a wavy line underneath. Below the title, there are three input fields: 1. "NAME OF DEAL" with a single-line text input box. 2. "GOAL WITH THE DEAL" with a dropdown menu showing "GENERATING SALES" and a checkmark icon. 3. "WHERE DO YOU WANT TO PROMOTE YOUR DEAL?" with two radio button options: "GLOBAL" and "SPECIFIC COUNTRIES". Below these, there is a question "WHICH CHANNELS SHOULD YOUR DEAL BE PROMOTED ON?" followed by three checkbox options: "INSTAGRAM", "YOUTUBE", and "TIKTOK". At the bottom of the form is a wide button labeled "NEXT".

PURPOSE OF THE DEAL

NAME OF DEAL

GOAL WITH THE DEAL

GENERATING SALES

WHERE DO YOU WANT TO PROMOTE YOUR DEAL?

☐ GLOBAL

☐ SPECIFIC COUNTRIES

WHICH CHANNELS SHOULD YOUR DEAL BE PROMOTED ON?

☐ INSTAGRAM

☐ YOUTUBE

☐ TIKTOK

NEXT

FIGURE 5.2: Close up on the "Purpose of deal" block of the deal creation process.

Figure 5.3 below shows a close up on the "Deal information" block of the deal creation process. The block contains all of the inputs listed above for this section.

DEAL INFORMATION

WHAT ARE YOU PROMOTING?

☐ PRODUCT

☐ SERVICE

WHAT TYPE OF PRODUCT/SERVICE?

ADD+ []

PRICE OF PRODUCT/SERVICE (OPTIONAL)

\$US []

TYPE OF COMMISSION FOR INFLUENCER

☒ PERCENTAGE OF SALES EARNINGS THROUGH LINK

☐ FIXED AMOUNT PER SALE

AMOUNT OF COMMISSION FOR INFLUENCER

[] 20%

DESCRIPTION

[]

ADD MEDIA (OPTIONAL)

+ []

NEXT

FIGURE 5.3: Close up on the "Deal information" block of the deal creation process.

Figure 5.4 below shows a close up on the choice of influencer block of the deal creation process. The section provides recommendations of influencers based on the input from the previous steps and information about the brand. The section also gives the opportunity for brands to search and filter all the influencers who has an active account at Splick.

CHOOSE AN INFLUENCER THAT IS
RIGHT FOR YOU AND YOUR DEAL

INFLUENCERS RECOMMENDED FOR YOU BY SPICK

NAME	LOCATION	CATEGORIES
☐ 	SWEDEN	FASHION      
☐ 	SWEDEN	     
☐ 	SWEDEN	     
☐ 	SWEDEN	     
☐ 		     
☐ 		     
☐ 		     

1 2 3 4 5

SEARCH INFLUENCERS

FILTER v

FIGURE 5.4: Close up on the choice of influencer block of the deal creation process.

5.4.2 The evaluation

The evaluation of the Lo-Fi prototype was conducted through a heuristic evaluation. The heuristic evaluation process is described under Section 3.2. The heuristic guidelines used are **Visibility of system status**, **User control and freedom**, **Recognition rather than recall**, **Aesthetic and minimalist design**, and **Help users recognize, diagnose and recover from errors**.

Visibility of system status

This heuristic guideline refers to the need for users to always be informed of system operations. Looking at Figure 5.1, the content of the deal creation process is divided into separate blocks. The separation makes it clear that the deal creation process requires different types of information, information about what the goal of the deal is, information about the deal itself, and finally choosing an influencer. Figure 5.1 also shows a progress bar highlighting each of the steps and shows which step the user is currently on. To make it more clear that each step must be completed in order it would be good to make the other blocks not currently in focus greyed out. Looking at the bottom of Figures 5.2 and 5.3 there is a button with the title "Next" that should be available to click once all the required information is inserted. This makes it easy for the user to know when it is time to move on to the next step in the process.

User control and freedom

User control and freedom refer to the possibility to move freely in a digital space where backward steps and undoing/redoin actions are possible. The user should be able to go back to each input and change the information. The process of returning to a block that is already completed is not displayed in the Lo-Fi prototype. There should either be a button to access the previous block if editing becomes unavailable once the "Next" button on each block is pressed or it should always be possible to scroll up to a previous block and edit it. Once the first two blocks are completed, Splick recommends influencers based on the information given through the use of machine learning. The user should still be able to go back and edit the previous sections. Since changing some of the inputs will alter which influencers are recommended the user should be informed about this. If the user chooses to edit any of the previous blocks the "Choosing an influencer" block should be made unavailable until the user is once again done inserting all of the necessary information.

Recognition rather than recall

Relevant instruction on how to use the system should be easily accessible and the system should minimize the cognitive load of the user. Looking at Figure 5.1 there is a small description of what an affiliate deal is that is fixed to the side of the screen during the deal creation process. Looking at Figures 5.2,

5.3, and 5.4 there are placeholders for a description of each block. Some of the inputs might be confusing for users new to the field of influencer marketing. There is no in-depth description of any of the inputs, to make sure that none of the users get confused filling out the required information there should be better descriptions of each input. To minimize the cognitive load of the user the section are divided into separate blocks containing different types of information. It is however unclear whether or not the separation is sufficient. The deal creation process could be separated into additional blocks to further minimize the cognitive load of the user.

Aesthetic and minimalist design

The Lo-Fi prototype does not include any unnecessary information that is not essential to the deal creation process. There are however some inputs that change the structure of the block. Certain choices during the deal creation process lead to further inputs being necessary for the user to fill out. For example when deciding to only target certain countries in Figure 5.2 the user needs to specify which countries, the input for this choice is not displayed in the Lo-Fi prototype. Should these optional inputs only show up when these choices have been made or could it be designed in a more intuitive way.

Help users recognize, diagnose and recover from errors.

There are few areas where the user can make an error. There are certain inputs that require a specific type of information, such as the price of a product/service being a numerical value. Should the user fail to insert a valid price the field where the error occurred should be highlighted and an explanation of the error should appear. As mentioned in the **user control and freedom** guideline the user might want to edit some aspects of the deal after Splick has recommended some influencers that suit their needs the user should be informed, perhaps through a popup, that the recommendations will disappear and be available once the user finishes the edits and clicks on the "Next" button.

Conclusion of heuristic evaluation

The specific heuristic principles for evaluating the Lo-Fi prototype were chosen to cover the entire design and try to find aspects that could be improved upon. More heuristic principles, either from the ten heuristic guidelines presented in Section 3.2 or from creating new guidelines, could be used but this may lead to redundancy due to repeated findings. The heuristic evaluation found problems and possible improvements regarding aspects to increase the usability of the process such as making future steps unavailable to edit and broader aspects such as further dividing the deal creation process into additional blocks to minimize the cognitive load of the user.

5.5 Hi-Fi prototype

The Hi-Fi prototype is based on the Lo-Fi prototype created in the previous iteration. Through the heuristic evaluation of the Lo-Fi conducted with the help of guidelines presented in Section 3.2, problems with the user interface were discovered along with possible solutions to make it more intuitive. This iteration tries to improve on the parts where the Lo-Fi prototype was lacking. The results of the Hi-Fi prototype consists of two parts, the design and an evaluation of the feedback given in the usability tests.

5.5.1 The design

The design made in Adobe XD is a more realistic version than the Lo-Fi prototype that a user can interact with and go through the entire deal creation flow. The Hi-Fi prototype is presented in Figure 5.5 - 5.13.

The screenshot displays the Splick web application interface. On the left is a green sidebar with the Splick logo and navigation links: Dashboard, Analytics, Influencers, Deals (highlighted), Settings, and Support. The top navigation bar is light gray, featuring a search bar labeled 'Search by influencer, category, or message', a 'Feedback?' link, an 'Invite influencer' button, and a user profile icon labeled 'Intam'. The main content area is titled 'Create a new deal' and includes a 'Back' link. Below the title is a progress indicator with four steps: 'Purpose of the deal' (active), 'Deal information', 'Deal content', and 'Choosing an influencer'. A 'Next' button is visible at the bottom of the form. The form itself is titled 'Purpose of the deal' and contains a text input for 'Name of the deal' and a section for 'What is your goal with the deal?' with two radio button options: 'Generating sales' and 'Publicity'. A 'Next' button is located at the bottom of the form.

FIGURE 5.5: Overview of the deal creation process and the "Purpose of the deal" block.

Purpose of the deal

General information about the deal and what you want to get out of it.

Name of the deal

Luxury watches

What is your goal with the deal?

☒ Generating sales ☐ Publicity

Generating sales

A deal aimed at generating sales drives visitors to your site and aims to achieve the maximum number of conversions. If you choose this type of campaign, we will match you with influencers best fit for these kinds of deals.

Next

FIGURE 5.6: "Purpose of the deal" block with filled in information about the deal.

Deal information

Information about the deal that you want to create, where and how do you want it to be marketed.

Where do you want to promote your deal?

☐ Global

☐ Specific countries

Which channels should your deal be promoted on?

☒ Tiktok ☒ Youtube ☒ Instagram

What are you promoting?

☐ Product

☐ Service

What type of product/service are you promoting?

Price of product/service (optional)

Type of commission for the influencer

☐ Percentage of sales earnings from customers spending

☐ Fixed amount per sale

FIGURE 5.7: "Deal information" block without any information filled in.

Deal information

Information about the deal that you want to create, where and how do you want it to be marketed.

Where do you want to promote your deal?

☐ Global

☒ Specific countries

Which countries do you want to promote your deal in?

Germany, Sweden, etc. ▼

Selected countries

 Sweden

Which channels should your deal be promoted on?

☐ Tiktok ☐ Youtube ☒ Instagram

What type of Instagram content do you want to promote?

☒ Instagram post

☐ Instagram story

What are you promoting?

☒ Product

☐ Service

What type of product are you promoting?

Watches Add +

Price of product (optional)

\$US ▼ 59.99

Type of commission for the influencer

☒ Percentage of sales earnings from customers spending

☐ Fixed amount per sale

Amount of commission

Specify the amount that the influencer should earn from the customers' spending through their marketing effort.

20 %

Next

FIGURE 5.8: "Deal information" block with filled in information about the deal.

Deal content

Information about the deal that you want to create, where and how do you want it to be marketed.

Description of the deal

Instructions for the influencer

Add media for the deal (optional)

You can add media for the deal with images of your product/service that the influencer can share on their social media platforms.

+

Next

FIGURE 5.9: "Deal content" block without any information filled in.

Deal content

Information about the deal that you want to create, where and how do you want it to be marketed.

Description of the deal

Deal promoting our new luxury watches that got released on December 3.

Instructions for the influencer

We want you to promote our deal through your social media.

1. Post the attached image as a new instagram post
2. Write in the description that your followers should go to our website through a link you've attached in your instagram bio.

Add media for the deal (optional)

You can add media for the deal with images of your product/service that the influencer can share on their social media platforms.



Next

FIGURE 5.10: "Deal content" block with filled in information about the deal.

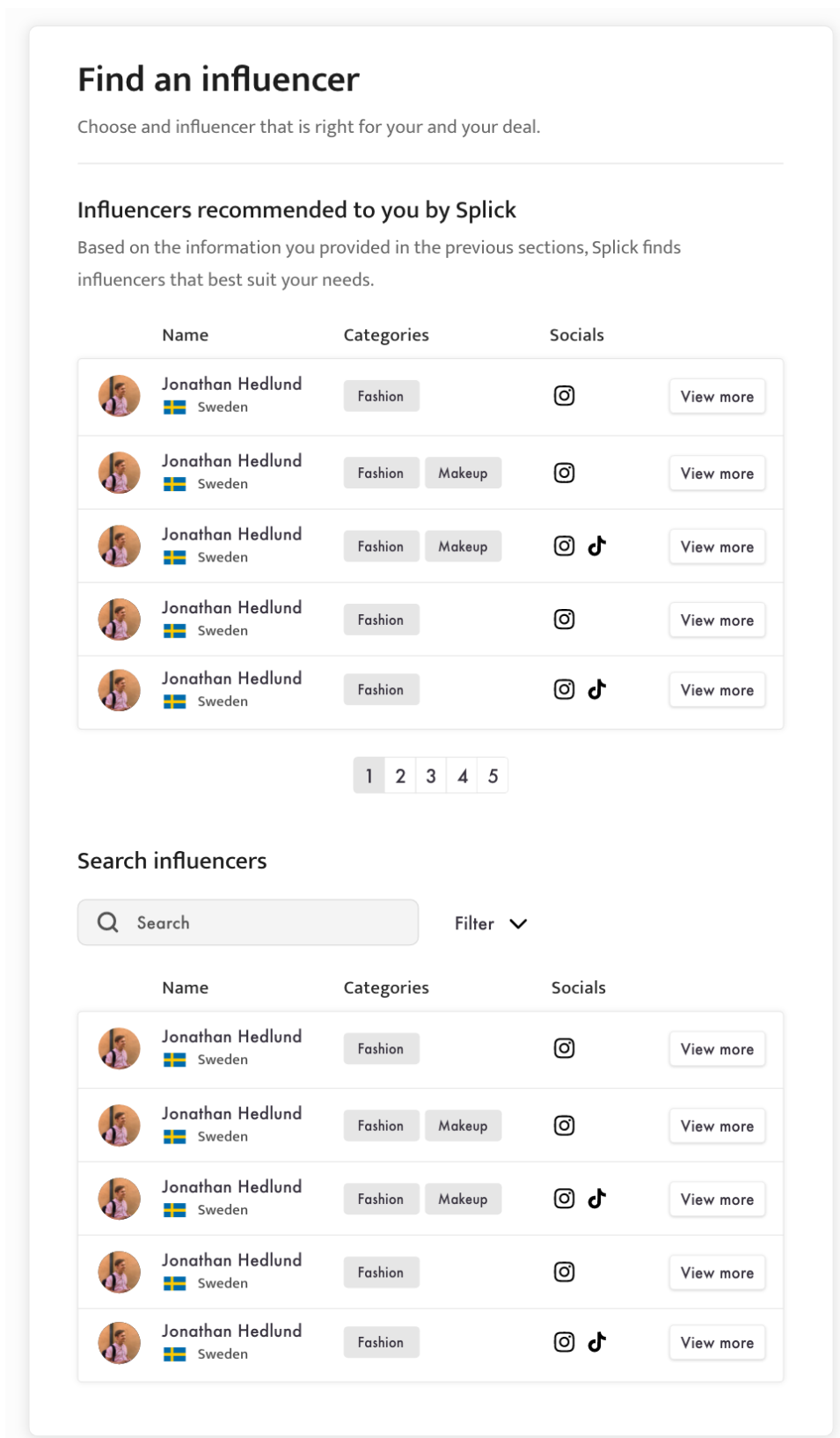


FIGURE 5.11: "Finding an influencer" section with influencers recommended by Splick's recommendation process and the option to search influencers on your own. Real influencers were displayed during the usability tests that the participant could choose.

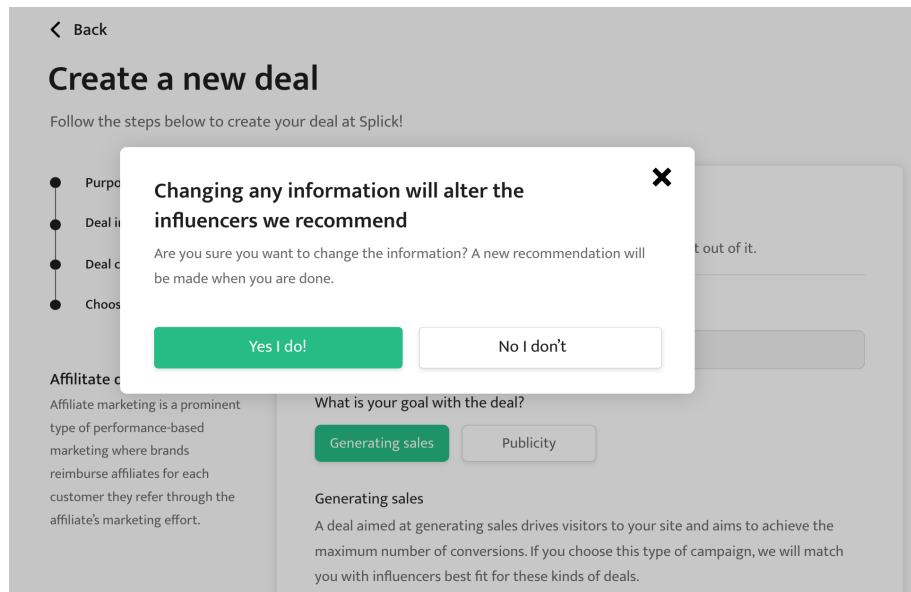


FIGURE 5.12: Popup for users when they try to change any of the information after influencers have been recommended to them based on their input.

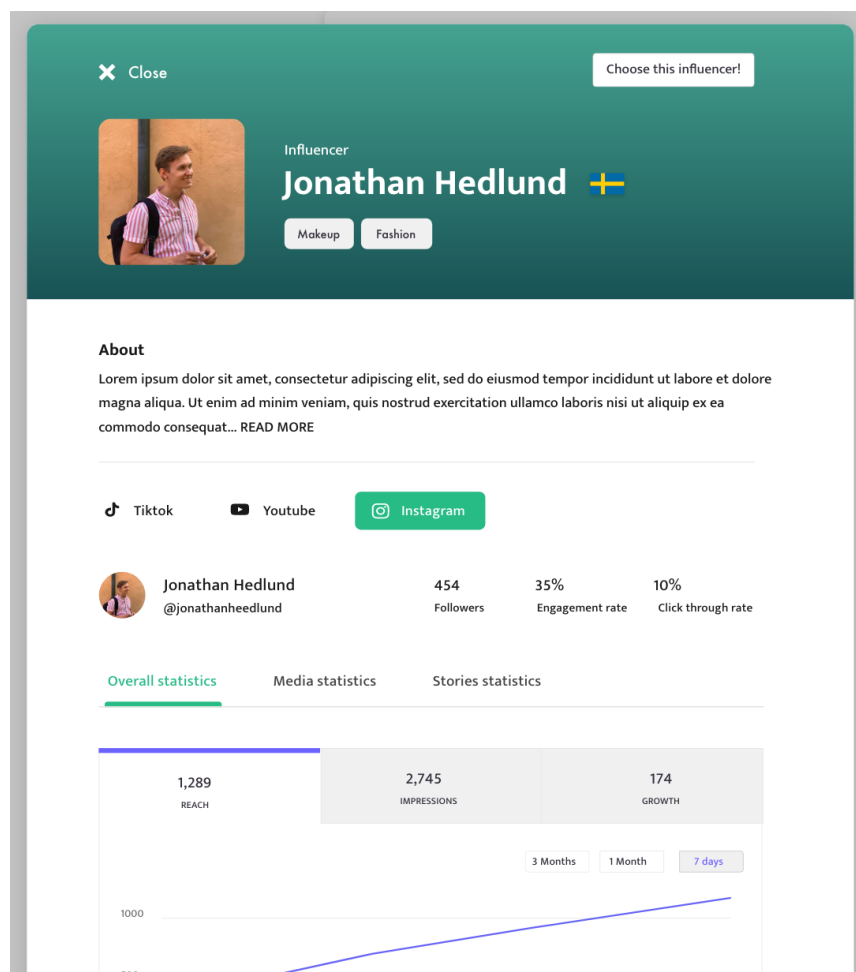


FIGURE 5.13: Popup to further inspect an influencer before deciding which one to cooperate with.

5.5.2 The evaluation

The tests were conducted through a zoom call where the participants shared their screen while interacting with the prototype. The facilitator gave an introduction to the field, a scenario they were put in, and read each task one by one as the test went on. The results of the usability tests are presented below.

1. *Define the purpose of the deal by giving it a name and choosing a deal with the goal of generating sales.*

All of the participants were able to complete this task without any problems. One participant mentioned that they appreciated the possibility of choosing between having a deal focused on "generating sales" and "publicity."

2. *You want to market your products in Sweden using Instagram posts and the influencer should be rewarded with a percentage rate of customers spending.*

The participants were able to complete this task without any major problems. Some of the participants noted that they were unsure if they could choose multiple social media platforms or not. One of the participants asked what they were supposed to do if the product that they were marketing is not listed in the "What type of product are you promoting?" dropdown in Figure 5.8.

3. *Provide all the information about the content of the deal before going to the next step.*

The participants noted that they were unsure of what they should write in the "Description of the deal" field and who they would write it for, themselves or the influencer. One participant noted that they would like to have the possibility to attach files to the deal and not just images. Some of the participants suggested that the final "Next" button before being recommended influencers should be named something along the lines of "Search influencers" or "Find an influencer to cooperate with" instead.

4. *Add Norway to the list of countries that you want to market your watches in.*

All but one participant scrolled up to the "Which countries do you want to promote your deal in?" field under "Deal information" which is the intended way to solve this task. The participant who did not do this right away tried to filter the searches under "Search influencers" shown in Figure 5.11.

5. *Search new influencers and pick one of the recommended influencers you want to cooperate with.*

All of the participants were able to solve this task with ease. One of the participants suggested that there should be a "Choose this influencer" button that is made more obvious in Figure 5.13.

6. *Additional thoughts the participants gave at the end of the test.*

The participants mentioned that they were not sure what a "Deal" meant. They thought the phrasing was too broad and that it should be more specific. Some of the participants noted some problems with the quality of life aspects related to the prototype. For example, they expected to be scrolled down automatically when pressing the "Next" button, something that the prototype did not deliver on.

Chapter 6

Discussion

This section will discuss the findings of the thesis and how this information can be used to further develop and research a solution for Splick's deal creation process. The specific topics of discussion are the input and output of the machine learning algorithm, suggestions of a suitable machine learning type, and the design process.

6.1 Input and output of the machine learning algorithm

All of the inputs mentioned in Section 2.5 are all interesting in their own way. The possibilities for combining and analyzing all the different metrics are immense. Depending on the deal and which type of brand and influencer are participating, the importance of each metric can vary. As mentioned in Section 5.1, the goal of the deal can vary depending on the brand's motive for the deal. If the deal's goal is to gain exposure, a high number of followers might be preferred. Or if the purpose is to sell a specific product/service, a combination of the number of followers, their click-through rate, and their engagement rate might be more important to look at. This shows how important it is to gather as much information about the brand and the deal as possible before looking at the social media benchmarks, similar deals that have happened in the past, and potential influencers.

If possible, it would be very interesting to provide as many metrics as possible to a machine learning algorithm to be able to see which connections are made and how the importance changed depending on what brand and what type of deal is being provided.

Whether or not a partnership is successful can be determined from a few factors, partly by Splick's benchmarks mentioned in Section 2.5.2 sales, add to carts, and clicks. But it could also be of benefit to provide the option for participants to rate their experience with the partnership. After a deal has been completed, both the brand and the influencer should be asked to rate their experience. This information could be used to further train the algorithm to find better matches in the future. The amount of sales, add to carts, and clicks all need to be evaluated related to the size of the influencer that participated in the deal. A machine learning solution would have to be taught to compare the results of partnerships related to the influencers. It

would then need to put more weight into updating it's decision making from partnerships with a higher relative success and the participants experience.

There are a lot of different metrics that are specific for each social media that could be researched in the future. There is a wide variety of social media platforms where influencers can reach out to their followers. The metrics mentioned in Section 2.5.1 have been gathered from looking at Instagram, Youtube, and TikTok. These social media platforms were chosen because they are the ones that Splick supports at the moment. Splick can gather data from users that have connected any of their accounts on those three platforms to Splick's website. The chosen metrics can be found on most of these social media platforms. But metrics on each social media are not equivalent to each other, for example it would not be wise giving the same weight to followers on Instagram and Youtube. It might be best to separate the metrics based on social media as well. This means for example that Instagram, Youtube, and Tiktok all get their own follower metric that the machine learning algorithm takes as input. This separation would make it possible for brands to sort which social media they want to promote on and are then provided with suitable influencers with a presence on that platform. The separation would make the match more accurate, but it would take longer to train since there are more metrics.

As mentioned in Section 5.2, the output for the algorithm could be points of interest that the algorithm notices when processing the input, which could be similarities in successful partnerships from previous cooperation with similar participants. It would be preferable to have the algorithm output a "type" of influencer by providing some key points that signify them. For example, the output could be, female influencers from Sweden with a follower size of 100,000+, and an engagement rate above 9%, and so forth. With the help of the output, it would make it possible to filter through Splick's catalog of influencers and provide a list of influencers meeting the provided criteria. This type of output would make it more clear what the similarities are in successful partnerships. It would move away from having a black box solution where the input gets processed in some magical way and then names of some influencers are provided.

With both the input and output of the machine learning algorithm, it will be necessary to review the outcome of an implemented solution. Reviewing the outcome will make it possible to iterate further and improve the system.

6.2 Suggestions regarding further development of a machine learning solution

As explained in Section 5.3.1, it would be difficult to implement a solution with the supervised learning approach since there would be a load of assumptions that need to be made manually for it to work. For example, what determines if the cooperation is successful. This is difficult since there is no clear formula that will work in all scenarios. Figuring out what works and

what does not work is one of the main points that we hope a machine learning solution can help figure out.

As stated in Section 5.3.2 and 5.3.3, reinforcement learning och unsupervised learning both seem to be viable approaches for Splick's recommendation system. Both can make discoveries on their own when provided with inputs. Unsupervised learning takes this input and finds associations between input values and groups them, while a reinforcement learning agent learn through delayed feedback after interacting with the environment it works in. Their approach differs but in this scenario they could both make great candidates for further research and development.

Whichever route is taken, it will take time to teach the algorithm which type of influencers to recommend. Reinforcement learning could be fine to use out of a data and time perspective since it can train and explore by evaluating data from all cooperation that takes place on Splick's website. It can be implemented and work in the background until we deem it to be competent enough to make recommendations on which influencers fit with which brands. But as explained in Section 5.3.3, it is hard to say how long that would take since it all depends on how many partnerships go down through Splick's services. As stated in Section 5.3.2, an unsupervised learning approach could be a good way to continue the development of a machine learning solution to solve Splick's problem. An unsupervised learning implementation could lower the time that it takes for competent recommendations to be given to the brand searching for a partner. Neural Network-based clustering seems like a promising way to go, as the algorithm receives more data the recommendations that it provides the brand with become more and more accurate.

With the time restraint of this master thesis, it was only able to scratch the surface of the different types of machine learning. The theory in Section 2.1 does not really go in-depth on the algorithms and the different types. This makes it hard to make decisions on which route would be the most preferable to take when starting development. But it does provide a good starting point. Further research will be needed in the field of unsupervised learning, partly researching the field more to learn more about the possibilities but also looking into more algorithms that might be better suited for Splick's recommendation system.

6.3 Design of the deal creation process

The design iterations developed and presented in Section 5.4 and 5.5 provided a solid foundation for Splick's deal creation process. As stated in Section 4.3.1, the design process consisted of two iterations, a Lo-Fi prototype, and a Hi-Fi prototype, both being evaluated although with different evaluation methods. The content of the process is partly based on the findings related to metrics a potential machine learning process could take into consideration stated in Section 6.1. Since it is not clear what makes a cooperation a successful one, it is necessary to include as many parameters as possible for

a machine learning algorithm to find correlations and provide suitable recommendations.

One limitation of the usability tests are that the participants were not the target group of the deal creation process. The participants were not brand managers or business owners looking to market their products through influencers, and they did not really have any prior knowledge in marketing through influencers. This means that the content of the deal creation could have been tested more optimally if the actual target group was used. Testing on the target group would for example result in insights about inputs that could have been missed that are necessary to include. I would however say that the usability tests were a success. Conducting usability tests with the participants that were used made it naturally focus more on the user experience throughout the process. Providing valuable insights about areas where it can be improved before moving on to the next iteration.

There is still a need for further work before finalizing a design. The feedback from the usability tests presented in Section 5.5.2 needs to be implemented in a third iteration of the deal creation process. After a third iteration of the design is completed, usability tests on the target group needs to be conducted to make sure that everything is included and works as expected before moving on to developing and implementing the design.

Chapter 7

Conclusion

7.1 Conclusion based on results and discussion

The objective of this master thesis was to provide a foundation for the deal creation process being implemented on Splick's website. The foundation was created through an extensive literature study to provide a path for conducting further research and development, and a design of the user interface for the deal creation process. The work of this thesis can be divided into four parts. First of all through examining different metrics related to brands, influencers, and the deal put out by the brand. These metrics can be used as input for a machine learning solution. Second, examining how the output of the algorithm should be designed in order to better recommend potential influencers to brands. Third, different machine learning types and algorithms were researched and analyzed in order to find a possible route for further research and development. Finally, a design of the deal creation process based on the findings related to metrics that are important for a machine learning algorithm to take into consideration.

The suggested input consists of metrics from the participants of the deal, information about the deal itself, metrics from the social media platforms where the influencer is active, and benchmarks from Splick's affiliate link service. The suggested output is common points of interest found in similar partnerships from the past that turned out to be more successful. The output is tailored to the type of deal and brand seeking to conduct business through Splick. Examples of possible outputs could be female influencers from a certain geographic location, with a follower size over a certain threshold, and so forth. This type of output would also make it more clear to the observer to actually be able to see which similarities there are in successful partnerships.

Unsupervised learning and reinforcement learning both seem to be able to be tasked with solving Splick's needs. Both of them can make discoveries on their own when provided with inputs. Although, unsupervised learning does seem to be superior regarding time and data volume in order to start making competent recommendations. So I suggest continued research into unsupervised learning as well as a Neural Network-based clustering approach. I hope that continued research into this branch of machine learning will result in a more clear plan for development and implementation.

The design of the deal creation process provides a good start. The evaluation of the different iterations resulted in valuable insight into existing problems and areas of improvement. Although, there is still a lot of work to be done before finalizing a design. Insights from Splick's target group is needed to find problems and receive suggestions to make it a better experience.

7.2 Future work

I believe that I was able to provide a solid foundation for Splick's deal creation process. Through the literature study, I was able to provide valuable insight into how the input and output of a machine learning algorithm should be designed. As well as a good idea on where to start conducting further research regarding which machine learning approach to take. There is still a need for testing different machine learning algorithms to be able to make a clear choice of which route to proceed with development. But to be able to perform tests with some different machine learning solutions I would have to provide data from previous partnerships between brands and influencers. Either from real partnerships or from made-up ones. Since Splick's services are not yet up and running I was not able to find real data that fit my requirements and would therefore need to make up test data. The design provides a clear and visual example of how a finished deal creation process could look like when implemented on Splick's website. Further work still needs to be done before finalizing a design, partly by implementing the feedback given from the participants during the usability tests but also through usability tests conducted on Splick's target group. I believe that with additional iterations of the design, the deal creation process will evolve into an integral part of Splick's service.

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